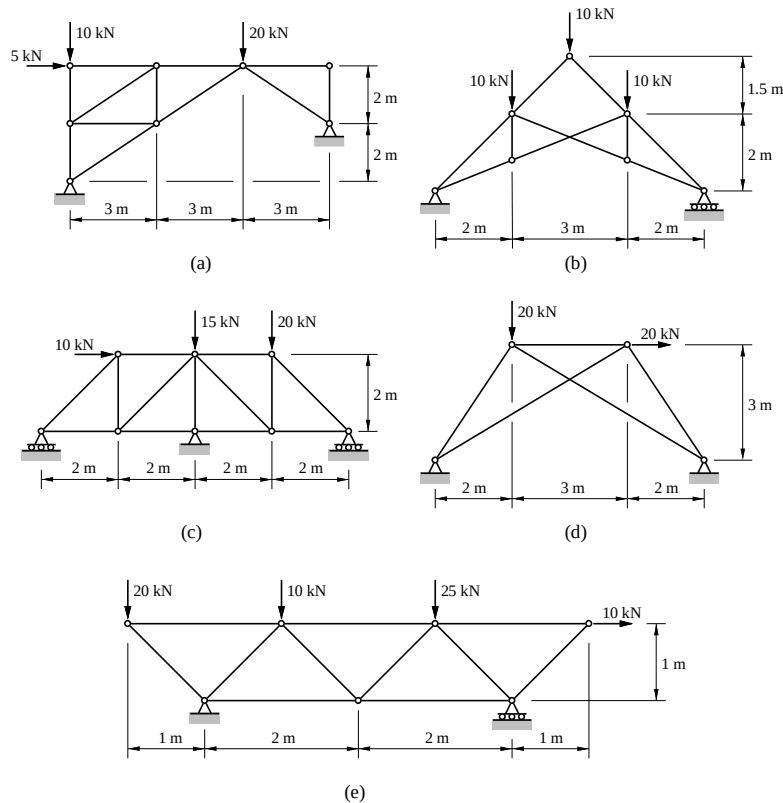
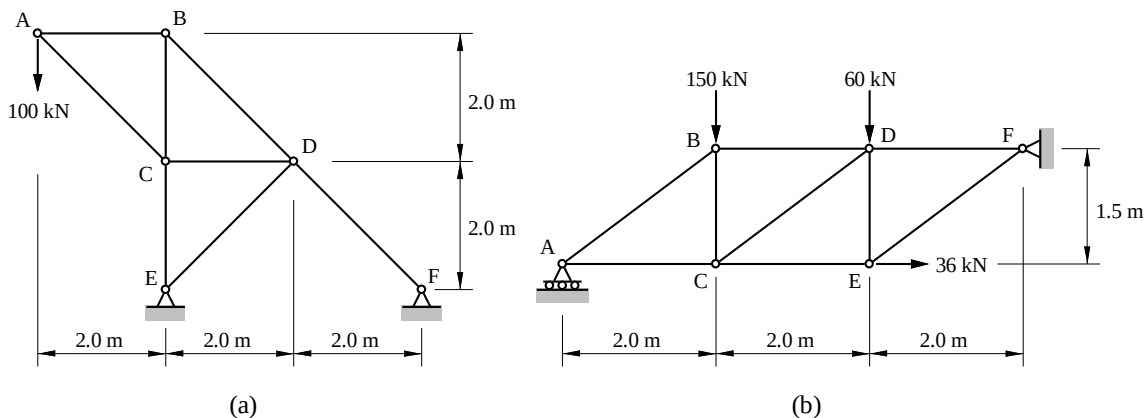


UNIVERSITY OF BIRMINGHAM
SCHOOL OF ENGINEERING
Mechanics-1: Tutorial 17– Truss Structures-1

1. The diagram shows a number of pin-jointed plane frames. Note that members cross without touching where no pins are shown. In each case, state whether the frame is statically determinate, indeterminate or a mechanism.



2. Use the Method of Joints to analyse each of the pin-jointed frames shown in the diagram.



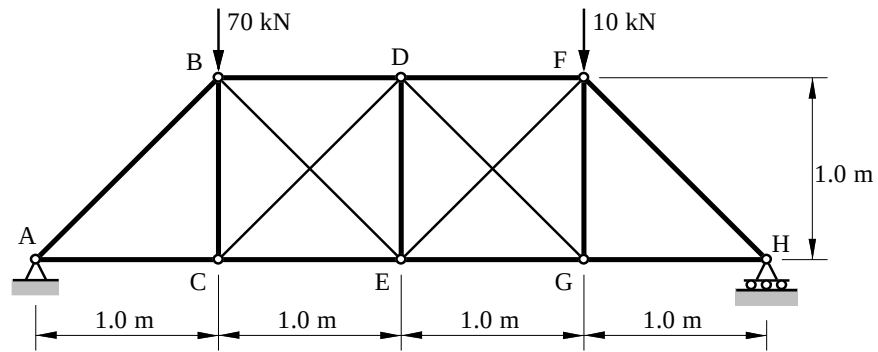
In each case, find the external reactions and the axial force in each of the members.

Ans :

	AB	AC	BC	BD	CD	CE	DE	DF	EF
1(a) (kN)	+100	-141.4	-100	+141.4	-100	-200	+70.7	+70.7	
1(b) (kN)	-215	+172	-21	-172	+35	+144	-81	-144	+135

3. In the pin-jointed frame shown in the diagram the cross-bracing members BE, CD, DG and EF are formed from steel wires, which are just taut when the frame is unloaded. Note that in each square panel only *one* wire will be extended, and will carry a tensile force - the other will be

shortened, and will go slack. All other members are solid bars which can carry tensile or compressive force.



What are the internal forces in the frame when vertical loads of 70 kN and 10 kN act at joints B and F respectively?

Ans :

	AB	AC	BC	BD	CE	DE	DF	EG	FG	FH	GH
(kN)	-77.8	+55	-15	-55	+40	-15	-40	+25	0	-35.4	+25

Answers to Tutorial Questions

1.	<u>Question</u>	<u>m</u>	<u>r</u>	<u>i</u>	<u>m+r</u>	<u>2i</u>	<u>Comment</u>
a		12	4	8	16	16	Statically determinate
b		10	3	7	13	14	Mechanism
c		13	4	8	17	16	Statically indeterminate
d		5	4	4	9	8	Statically indeterminate
e		11	3	7	14	14	Statically determinate

2a Resolve horizontally:

$$H_E + H_F = 0, \Rightarrow H_E = -H_F$$

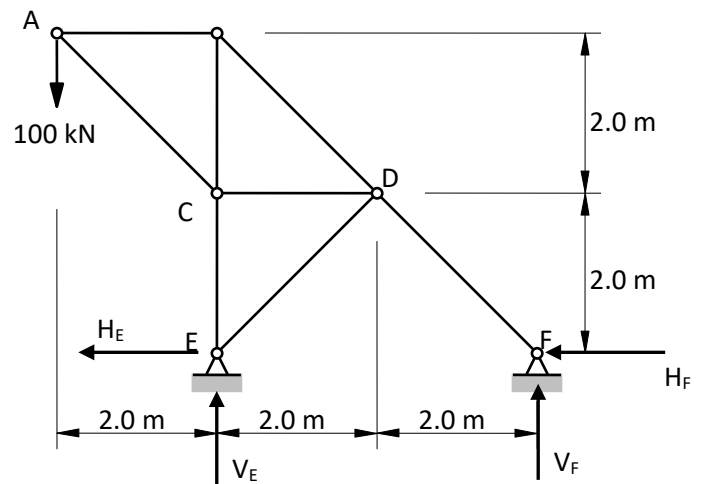
Resolve vertically:

$$V_E + V_F - 100 = 0 \Rightarrow V_E + V_F = 100$$

Take moments about E (+ve clockwise)

$$-(100 \times 2) - (4V_F) = 0 \Rightarrow V_F = -50 \text{ kN}$$

$$\therefore V_E = 150 \text{ kN}$$



Consider Joint F

$$\text{Resolve vertically: } T_{DF} \cos 45 + V_F = 0 \quad T_{DF} = -(-50) \sqrt{2} = 70.7 \text{ kN}$$

$$\text{Resolve horizontally } T_{DF} \cos 45 + H_F = 0 \quad H_F = -50 \text{ kN}$$

Using method of joints, move through joints in the order A, B, C and then D calculating the member forces. Check your answers at joint E.

2b Resolve vertically:

$$V_A + V_F - 150 - 60 = 0$$

$$V_A + V_F = 210$$

Resolve horizontally:

$$H_F = 36 \text{ kN}$$

Taking moments about F (+ve clockwise)

$$(V_A \times 6) - (150 \times 4) - (60 \times 2) - (36 \times 1.5) = 0 \quad \Rightarrow V_A = 129 \text{ kN, hence}$$

$$V_F = 81 \text{ kN}$$

Using the method of joints, move through the joints in the order A, B, C, E and D calculating the member forces. Check at joint F.

3 Resolve vertically: $V_A + V_H - 70 - 10 = 0$. $V_A + V_H = 80$

Resolve horizontally: $H_A = 0$

Take moments about point A (+ve anticlockwise)

$$(70 \times 1) + (10 \times 3) - (4V_H) = 0. \quad \Rightarrow V_H = 25 \text{ kN, hence } V_A = 55 \text{ kN.}$$

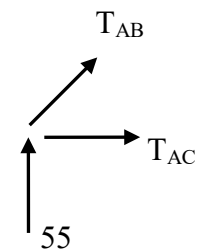
Consider joint A

Resolve vertically: $T_{AB} \cos 45 + 55 = 0$

$$\therefore T_{AB} = -77.8 \text{ kN (compression)}$$

Resolve horizontally: $T_{AB} \cos 45 + T_{AC} = 0$

$$\therefore T_{AC} = 55 \text{ kN (tension)}$$



Assumption 1: CD is in tension. Hence, no force exists in BE and this member can be ignored.

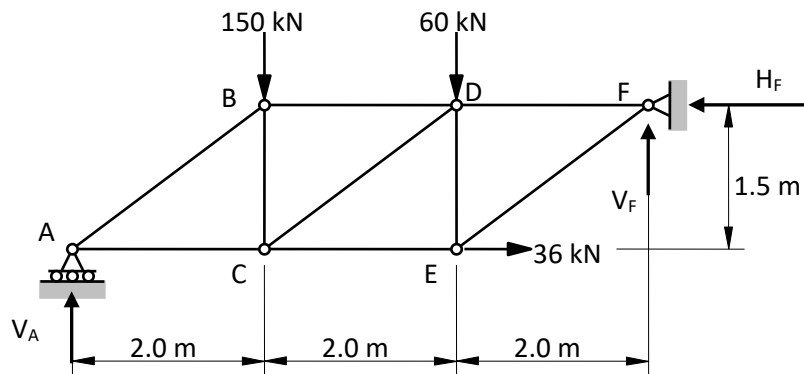
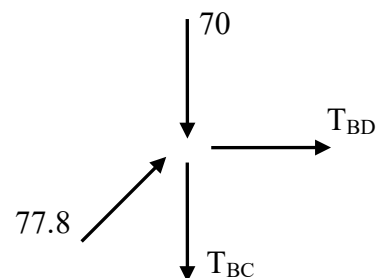
Consider joint B

Resolve vertically: $77.8 \cos 45 - 70 - T_{BC} = 0$

$$\therefore T_{BC} = -15.0 \text{ kN (compression)}$$

Resolve horizontally: $77.8 \cos 45 + T_{BD} = 0$

$$\therefore T_{BD} = -55 \text{ kN (compression)}$$



Consider joint C

Resolve vertically: $T_{CD}\cos 45 - 15.0 = 0$

$$\therefore T_{CD} = 21.2\text{kN (tension)}$$

Hence assumption 1 is correct.

Resolve horizontally: $21.2\cos 45 + T_{CE} - 50 = 0$

$$\therefore T_{CE} = 40.0\text{kN (tension)}$$

Assumption 2: EF is in tension. Hence, no force exists in DG and this member can be ignored.

Consider joint D

Resolve vertically: $-T_{DE} - 21.2\cos 45 = 0$

$$\therefore T_{DE} = -15.0\text{kN (compression)}$$

Resolve horizontally: $T_{DF}\cos 45 - 15 = 0$

$$\therefore T_{DF} = -40.0\text{kN (compression)}$$

Consider joint E

Resolve vertically: $T_{EF}\cos 45 - 15 = 0$

$$\therefore T_{EF} = 21.2\text{kN (tension)}$$

Hence, assumption 2 is correct

Resolve horizontally: $21.2\cos 45 + T_{EG} - 45 = 0$

$$\therefore T_{EG} = 25.0\text{kN (tension)}$$

Consider joint F

Resolve horizontally: $40 + T_{FH}\cos 45 - 21.2\cos 45 = 0$

$$\therefore T_{FH} = -35.4\text{kN (compression)}$$

Resolve vertically: $-10 - 21.2\cos 45 + T_{FG} - (-35.4)\cos 45 = 0$

$$\therefore T_{FG} = 0$$

Consider joint G

Resolve horizontally: $T_{GH} - 25.0 = 0$

$$\therefore T_{GH} = 25.0\text{kN (tension)}$$

Check at joint H:

Resolve horizontally: $25.0 - 35.4\cos 45 = 0 \quad \checkmark$

Resolve vertically: $25.0 - 35.4\cos 45 = 0 \quad \checkmark$

